Individual Contribution Report
CMPE321 Project 3

Person A

Individual Contributions

Disk Space Manager (DSM)

- Implemented raw page I/O using `seek()` for direct page access
- Built IO counter to track disk read/write operations
- Designed page file structure for all record types
- Implemented `log_write` stub for operation logging

Buffer Manager (BM)

- Implemented buffer pool using `OrderedDict` for O(1) eviction tracking
- Built LRU eviction policy (evict least recently used page)
- Built MRU eviction policy (evict most recently used page)
- Integrated buffer pool with DSM for transparent page caching

Experiment Videos

- Designed and recorded experiment: LRU vs. MRU buffer policy comparison
- Designed and recorded experiment: Index strategy comparison (heap scan, hash index, B+ tree)
- Designed and recorded experiment: Buffer pool size sensitivity analysis

Collaboration

Both team members actively collaborated throughout the project. We began by exchanging ideas on the overall system design, discussing architectural decisions such as page format, buffer eviction policies, and index strategy trade-offs before writing any code. Once implementation was underway, we reviewed each other's code to ensure correctness and consistency across layers. Finally, we tested the complete system together, verifying all commands and index strategies against the provided sample input as well as our own generated workloads.

Self-Assessment

Working on the lower layers of the DBMS gave me a deeper understanding of how data is physically stored and retrieved on disk, and how buffer management bridges raw I/O with higher-level operations. Implementing both LRU and MRU policies made clear how eviction strategy directly impacts performance depending on access patterns. Designing and running the three experiments also strengthened my ability to isolate variables and interpret system-level performance results in a meaningful way. An area for improvement is anticipating edge cases earlier in the design phase; during testing, several corner cases around page eviction under a full buffer pool required revisiting the implementation. Overall, this project gave me practical experience with concepts that are often only described theoretically in lectures.

Use of AI Tools

Claude (Anthropic) was used as an AI assistant during the development of the DSM and BM components. It was primarily used to clarify design decisions, such as the trade-offs between different eviction strategies and the appropriate use of Python's `OrderedDict` for maintaining access order efficiently. For the experiment design, Claude was used to help structure the workloads and think through what variables to control across each test scenario. AI assistance helped produce cleaner and more consistent implementations by suggesting edge cases to handle and pointing out potential issues before they surfaced during testing. All final code and analysis decisions were made and verified by the team.